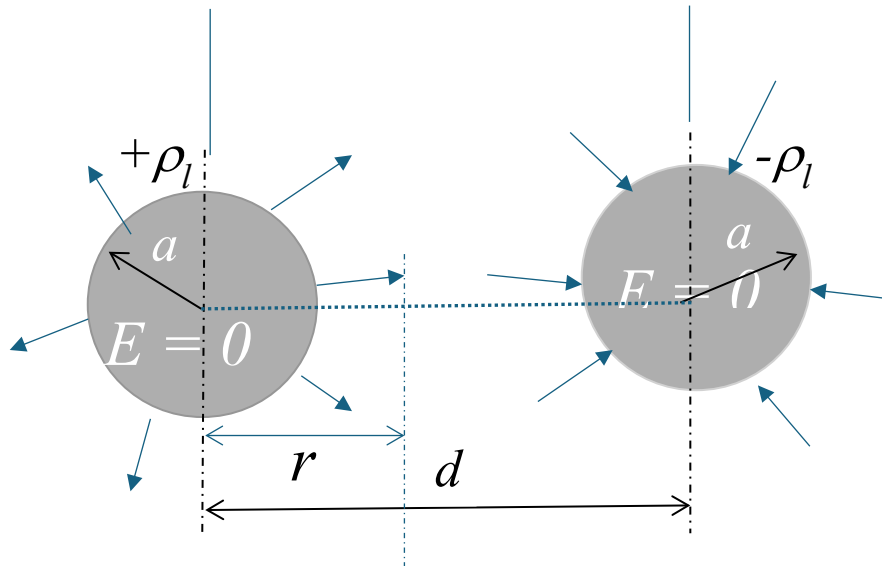


Solutions 2025

Question 1

(i)

We assume a line charge density ρ_l in the conductors, one with positive polarity and the other with negative polarity. To apply Gauss's law, we must also assume that $d \gg a$, so that the electric field lines in the Gaussian surface around the conductors are normal, even in the space between the two conductors.



Thus, the two electric fields of the two conductors are at an arbitrary radial distance r from the centre of each conductor:

$$\oint_s \vec{E}_1 \cdot d\vec{s} = \frac{Q_{\text{enclosed}}}{\epsilon_0} \rightarrow 2\pi r L |E_1| \vec{a}_{r1} = \frac{\rho_l L}{\epsilon_0} \rightarrow \vec{E}_1 = \frac{\rho_l}{2\pi\epsilon_0 r} \vec{a}_{r1}$$

Similarly, for the negative polarity conductor,

$$\vec{E}_2 = \frac{\rho_l}{2\pi\epsilon_0 r} \vec{a}_{r2}$$

Unit vector a_{r1} represents outward radial direction, whereas a_{r2} represents inward radial direction i.e., $a_{r1} = -a_{r2} = a_r$.

Thus,

$$\vec{E}_1 = \frac{\rho_l}{2\pi\epsilon_0 r} \vec{a}_r, \quad \vec{E}_2 = \frac{-\rho_l}{2\pi\epsilon_0 r} \vec{a}_r$$

Potential at conductor 1 due to $+\rho_l$ considering the other conductor as the reference,

$$V_1 = - \int_{d-a}^a \vec{E}_1 \cdot d\vec{r} = - \int_{d-a}^a \left(\frac{\rho_l}{2\pi\epsilon_0 r} \vec{a}_r \right) \cdot (dr \vec{a}_r) = \left[\frac{\rho_l}{2\pi\epsilon_0} \ln \frac{d-a}{a} \right] \text{ Volt}$$

Similarly, potential at the conductor 2 due to $-\rho_l$,

$$V_2 = - \int_{d-a}^a \vec{E}_2 \cdot d\vec{r} = - \int_{d-a}^a \left(\frac{-\rho_l}{2\pi\epsilon_0 r} \vec{a}_r \right) \cdot (dr \vec{a}_r) = \left[\frac{-\rho_l}{2\pi\epsilon_0} \ln \frac{d-a}{a} \right] \text{ Volt}$$

The potential difference between the two conductors:

$$\therefore V_{12} = V_1 - V_2 = \left(\frac{\rho_l}{\pi\epsilon_0} \ln \frac{d-a}{a} \right) \text{ Volt}$$

2 marks

(For full marks, must state assumptions and show derivation of at least one field)

2 marks

(For full marks, must show derivation of at least one potential and the potential between two conductors V12.)

Capacitance per meter [F/m] between the two conductors:

$$\therefore C = \frac{\rho_l}{V_{12}} = \frac{\pi\epsilon_0}{\ln \frac{d-a}{a}} \text{ F/m, If } d \gg a, C \approx \frac{\pi\epsilon_0}{\ln \frac{d}{a}} \text{ F/m}$$

1 mark

Using the numerical values,

$$C = \frac{\pi \times 8.854 \times 10^{-12}}{\ln \left(\frac{0.65 - 2.5 \times 10^{-3}}{2.5 \times 10^{-3}} \right)} \text{ F/m}$$

$$= \frac{2.78 \times 10^{-11}}{5.5568} = 0.5 \times 10^{-11} \text{ F/m}$$

$$= 5 \text{ pF/m}$$

OR

$$C = \frac{\pi \times 8.854 \times 10^{-12}}{\ln \left(\frac{0.65}{2.5 \times 10^{-3}} \right)} \text{ F/m}$$

$$= \frac{2.78 \times 10^{-11}}{5.56} \text{ F/m}$$

$$= 0.5 \times 10^{-11} \text{ F/m}$$

$$= 5 \text{ pF/m}$$

2 marks

(ii) The lossy resistance can be calculated easily using the relation between the capacitance and resistance of a lossy medium,

$$RC = \frac{\epsilon}{\sigma} \rightarrow R = \frac{\epsilon}{\sigma} \frac{1}{C} [\Omega/\text{m}] = \frac{\epsilon_0}{\sigma} \frac{1}{C} [\Omega/\text{m}] = \frac{8.854 \times 10^{-12}}{1.4 \times 10^{-15}} \frac{1}{5 \times 10^{-12}} [\Omega/\text{m}] = 1.26 \times 10^{15} [\Omega/\text{m}]$$

3 marks

(2ii)

1 mark

The potential between the two conductors $V_{12} = \frac{q_l}{\pi \epsilon_0} \ln\left(\frac{d-a}{a}\right)$ as derived in (i)

—(1)

From Gauss's law, the electric field, $\vec{E} = \frac{q_l}{2\pi\epsilon_0 r} \vec{a}_r$ —(2)

From (1),

1 mark

$$q_l = \frac{V_{12}(\pi\epsilon_0)}{\ln\left(\frac{d-a}{a}\right)}$$

Substituting in (2),

1.5 marks

$$\vec{E} = \frac{V_{12}(\pi\epsilon_0)}{\ln\left(\frac{d-a}{a}\right)} \cdot \frac{\vec{a}_r}{2\pi\epsilon_0 r}$$

$$= \frac{V_{12}}{2 \ln\left(\frac{d-a}{a}\right) r} \vec{a}_r$$

1.5 marks

When $r = a$, we will get maximum electric field (E_{max})

$$|E_{max}| = \frac{V_{12}}{2 \ln\left(\frac{d-a}{a}\right) \cdot a}$$

Considering $|E_{max}| = 0.50 \times 3 \text{ MV/m} = 1.5 \text{ MV/m}$, maximum

voltage can be calculated,

2 marks

$$V_{max} = |E_{max}| \left(2 \ln\left(\frac{d-a}{a}\right) \cdot a \right) \text{ V}$$

$$= 1.5 \times 10^6 \left[2 \ln\left(\frac{0.65 - 2.5 \times 10^{-3}}{2.5 \times 10^{-3}}\right) \times 2.5 \times 10^{-3} \right] \text{ V}$$

$$= 1.5 \times 10^6 \times 2 \times 5.5568 \times 2.5 \times 10^{-3} \text{ V} = 41.676 \text{ kV}$$

(iv)

When applying Gauss's law to calculate capacitance between two conductors, we assumed $d \gg a$, which implies that the electric fields of the two conductors are independent of each other; the field of one does not influence the field of the other. This assumption is not valid if the distance between the conductors is shorter.

2 marks (key words to get full marks are highlighted.)

Due to the influence of the electric field of one conductor on the other, the field lines may not remain normal on the assumed cylindrical Gaussian surface around the conductor. The line charge density of each conductor may also not remain uniform. In such situations, the method of image can be used to calculate the potential on each conductor, considering the line charge density of one conductor to be the image of the other conductor's charge density. In the method of image, the potential of each conductor is then the net potential of two line charge densities. Now that we consider the influence of both conductors when calculating the potentials, the capacitance value can also be calculated more accurately.

Question 2

(a)

The capacitance between the conductor 1 and the shield conductor 2 can be found using Gauss's law.

E at any radial distance r from the surface of the conductor within the dielectric i.e., $r_1 < r < r_2$,

$$\vec{E} = \vec{a}_r \frac{\rho_l}{2\pi\epsilon r} \text{ V/m, where } r_1 < r < r_2$$

Potential V between the conductor and the shield,

$$V_{ab} = -\int_{r_2}^{r_1} \vec{E} \cdot d\vec{r} = -\int_{r_2}^{r_1} \left(\vec{a}_r \frac{\rho_l}{2\pi\epsilon r} \right) \cdot (\vec{a}_r dr) = \frac{\rho_l}{2\pi\epsilon} \ln\left(\frac{r_2}{r_1}\right)$$

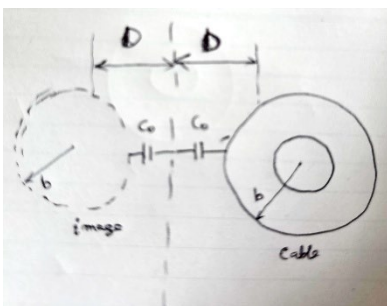
Capacitance per unit length:

$$C_{12} = \frac{\rho_l}{V_{ab}} = \frac{2\pi\epsilon}{\ln\left(\frac{r_2}{r_1}\right)} \text{ [F/m]} = \frac{2\pi \times 3\epsilon_0}{\ln\left(\frac{5}{1}\right)} \text{ [F/m]} = 103.7 \text{ [pF/m]}$$

optional

3 marks

The capacitance between the shield conductor and the wall can be found using the method of image.



4 marks

$$C = \frac{C_{20}}{2} = \frac{\pi\epsilon}{\ln\left[\left(\frac{d}{2a}\right) + \sqrt{\left(\frac{d}{2a}\right)^2 - 1}\right]} \text{ F/m, } d = 2D, a = r_2$$

$$\begin{aligned} C_{20} = 2C &= \frac{2\pi(3\epsilon_0)}{\ln\left[\left(\frac{2D}{2r_2}\right) + \sqrt{\left(\frac{2D}{2r_2}\right)^2 - 1}\right]} = \frac{6\pi\epsilon_0}{\ln\left[\left(\frac{D}{r_2}\right) + \sqrt{\left(\frac{D}{r_2}\right)^2 - 1}\right]} \\ &= \frac{6\pi \times 8.854 \times 10^{-12}}{\ln\left[\left(\frac{20}{5}\right) + \sqrt{\left(\frac{20}{5}\right)^2 - 1}\right]} = \frac{166.894 \times 10^{-12}}{\ln(4 + \sqrt{15})} = 80.88 \text{ [pF/m]} \end{aligned}$$

(b) When the shield conductor is grounded, it becomes a negatively charged body, as shown in the figure below:

If the outside body C is positively charged, the capacitance between B and C can form but not between C and A. If the C is negatively charged and at the same potential as the B, then there will not be any capacitance between B and C.

At initial condition, no charge or voltage in A,

$$Q_A = C_A V_A + C_{AB} (V_A - V_B) + C_{AC} (V_A - V_C)$$

$$0 = C_A 0 + C_{AB} (0 - V_B) + C_{AC} (0 - V_C) \rightarrow C_{AC} = C_{AB} \frac{V_B}{V_C}$$

If B is grounded, it will be at zero potential and C_{AC} will be zero.

Later, when the body A is charged,

$$Q_A = C_A V_A + C_{AB} (V_A - V_B), V_B = 0$$

In the case of the cable and the wall, the C represents the wall and A the inner conductor and B is the shield.

2 marks

(c)

Conductor 1 is at V_0 potential and conductor 2 at zero potential.

Using Laplace's equation for cylindrical coordinate,

$$\nabla^2 V = 0$$

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 V}{\partial \theta^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

The potential varies only in the radial direction.

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{dV}{dr} \right) = 0$$

It can be solved to find a potential function by direct integration,

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{dV}{dr} \right) = 0$$

Integrating once,

$$r \frac{dV}{dr} = k_1 \rightarrow \frac{dV}{dr} = \frac{k_1}{r}$$

Integrating again,

$$V(r) = k_1 \ln(r) + k_2$$

Integration constants k_1 and k_2 can be found from the boundary conditions,

$$V(r = r_1) = V_0 \text{ and } V(r = r_2) = 0$$

$$V(r_1) = k_1 \ln(r_1) + k_2 = V_0,$$

$$V(r_2) = k_1 \ln(r_2) + k_2 = 0$$

Solving for k_1 and k_2 ,

$$k_1 = \frac{V_0}{\ln\left(\frac{r_1}{r_2}\right)} \quad \text{and} \quad k_2 = -k_1 \ln(r_2) = \frac{-V_0 \ln(r_2)}{\ln\left(\frac{r_1}{r_2}\right)}$$

$$V(r) = k_1 \ln(r) + k_2$$

$$V(r) = \frac{V_0}{\ln\left(\frac{r_2}{r_1}\right)} [\ln(r) - \ln(r_2)], \quad r_1 \leq r \leq r_2$$

Electric field,

$$\vec{E}_r = -\vec{\nabla} V(r)$$

$$\vec{E}_r = -\vec{a}_r \frac{d}{dr} \left(\frac{V_0}{\ln\left(\frac{r_2}{r_1}\right)} (\ln(r) - \ln(r_2)) \right) = -\vec{a}_r \frac{V_0}{r \ln\left(\frac{r_2}{r_1}\right)}$$

(d)

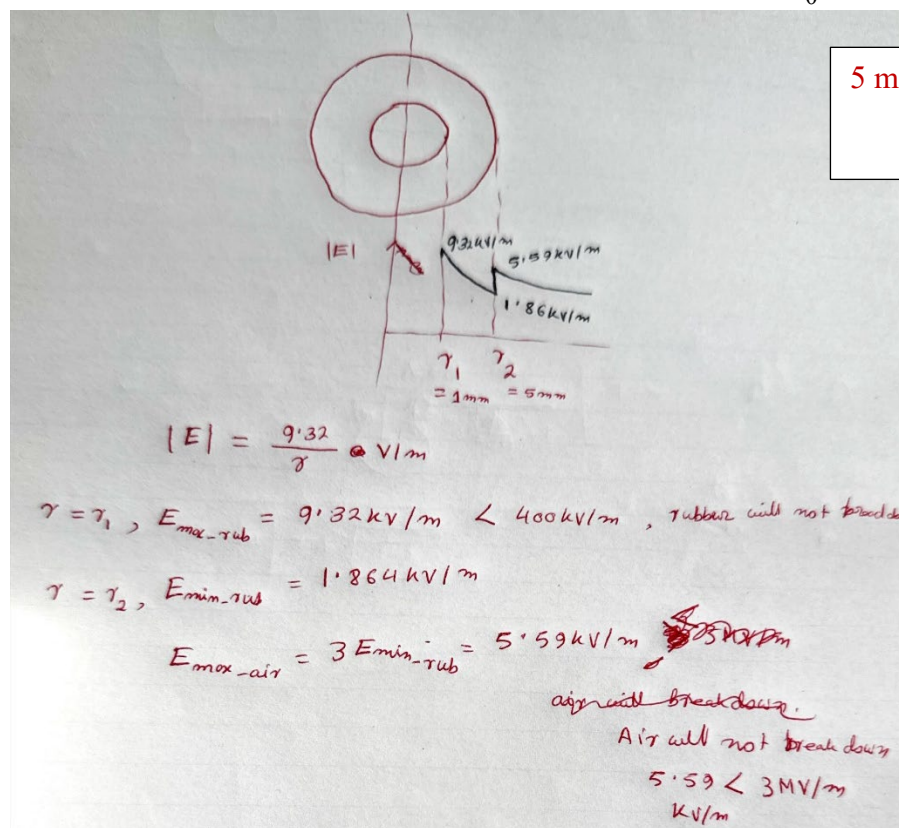
The electric field is maximum for the rubber when $r = r_1$ and for air when $r = r_2$.

$$\vec{E}_r = -\vec{a}_r \frac{V_0}{r \ln\left(\frac{r_2}{r_1}\right)} = -\frac{9.32}{r} \vec{a}_r \text{ [V/m]}$$

At the air and rubber interface, $r = r_2$,

$$|D_{n_1}| = |D_{n_2}| \rightarrow \epsilon_{rub} E_{min_rub} = \epsilon_0 E_{max_air} \rightarrow E_{max_air} = \frac{\epsilon_{rub} = 3\epsilon_0}{\epsilon_0} E_{min_rub} = 3E_{min_rub}$$

7 marks

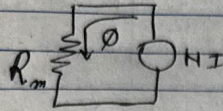


5 marks

Question 3

Question 3

(i)



4 marks

$$\Phi = \frac{NI}{R_m}, \quad R_m = \frac{l_m}{\mu_0 \mu_r S_m}$$

$$B_m = 1.28 \text{ T}$$

$$B_m S_m = \frac{NI \mu_0 \mu_r}{l_m}$$

$$\Rightarrow I = \frac{B_m l_m}{\mu_0 N} = \frac{1.28 \times 8 \times 10^{-3}}{4\pi \times 10^{-7} \times 1000} = 8.15 \text{ A}$$

(ii) With the coil current zero, when Ampere's law is applied to the structure

$$\oint \vec{H} \cdot d\vec{l} = I_{enc} = NI$$

$$H_m l_m = 0$$

$$H_m = 0$$

3 marks

From the B-H characteristic, when $H=0$, $B=1.25 \text{ T}$

(iii) With the air gap,

$$H_m l_m + H_c l_c = 0$$

$$H_m l_m = -H_c l_c$$

$$\frac{B_m}{\mu_0} l_m = -H_c l_c$$

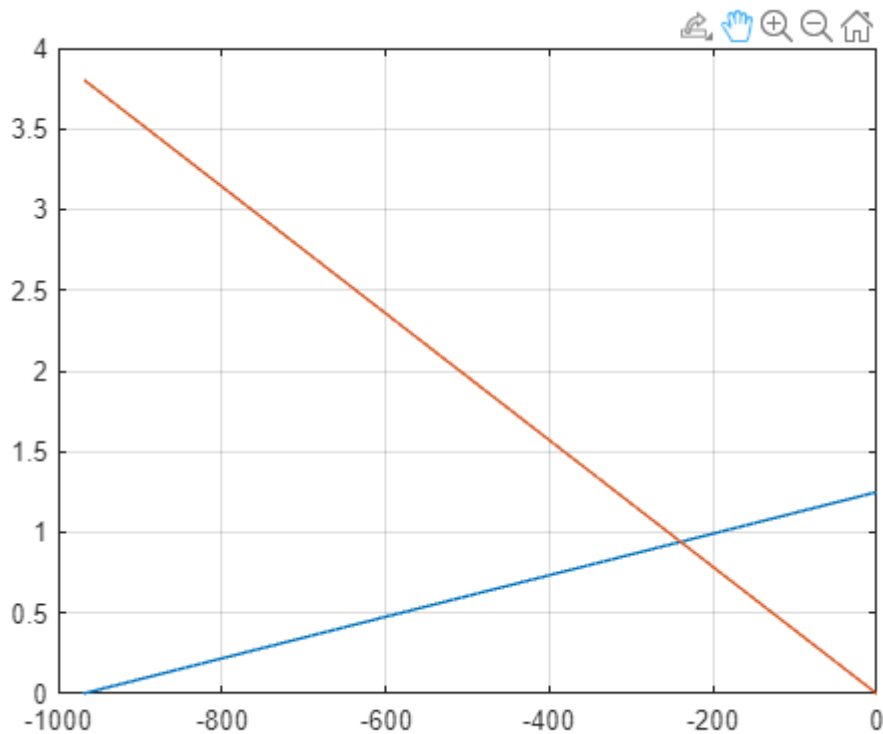
$$B_m = -\frac{\mu_0 l_c}{l_m} H_c \Rightarrow y = -mx$$

$$\text{slope of the load line} = \frac{\mu_0 l_c}{l_m} = \frac{4\pi \times 10^{-7} \times 25}{8}$$

$$B = 3.92 \times 10^{-6} \text{ T}$$

3 marks

The students will be given graph paper to plot the two graphs:



Red line: load line, blue line: magnet B-H curve.

Magnet $H=[0, -940\text{kA/m}]$; $B=[1.25, 0]$

Load line $H=[0, -940\text{kA/m}]$; $B=[0, 3.81]$

Load line B value is calculated using $B = -\mu_0 H, m = \frac{\mu_0 l_c}{l_m} = 3.92 \times 10^{-6}$

4 marks

From the graph, the value of the intersecting point, i.e., the magnet's operating points:

$B_m=0.94\text{T}$, $H_m=-239.6\text{kA/m}$

From the graph of load line, $B_m = 0.941\text{T}$
 $H_m = -239.62\text{ kA/m}$

(iii)

$$\text{Energy} = \left(\frac{1}{2} B_m H_m \right) \times \text{Volume of magnet}$$

$$= \frac{1}{2} \times 0.941 \times (239.62) \times (l_m \times a \times d)$$

$$= \frac{1}{2} B_g^2 \times \text{Volume at the air gap}$$

$$= \frac{1}{2} \times (0.4705)^2 \times (c \times d \times l_c)$$

$$B_0 S_m = B_g S_g$$

$$B_g = B_0 \frac{S_m}{S_g}$$

$$B_g = 0.941 \times \frac{a \times d}{c \times d}$$

$$= 0.4705\text{T}$$

6 marks

These energies have come from the electrical energy (i.e., the current in the coil) used to magnetise the magnet.